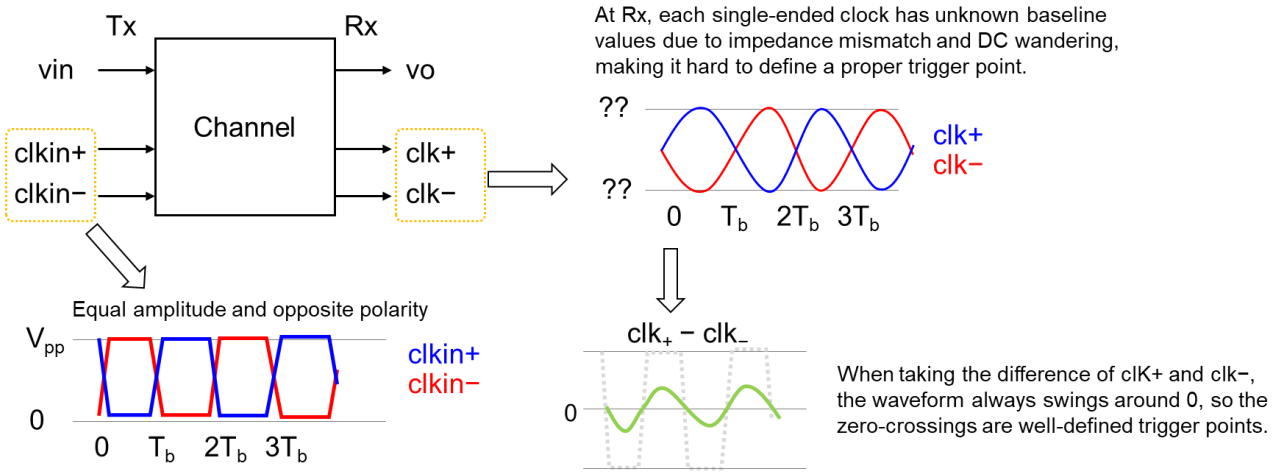
Homework: Construction of statistical eye – fixed-rate and triggered

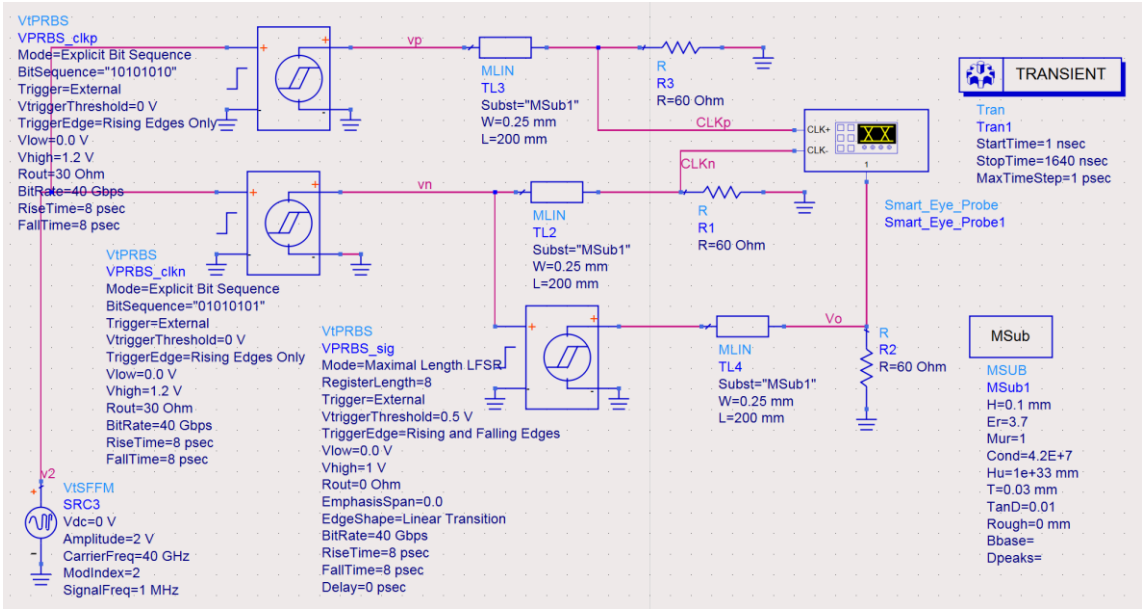
A signal is transmitted from Tx (`vin`) to Rx (`vo`). Its corresponding clock is a "differential clock", meaning that two lanes are dedicated to transmitting the clock, with opposite polarity (see figure below). The Rx looks at their difference: $\texttt{clk} = \texttt{clk}_+ - \texttt{clk}_-$; the trigger points are the zero-crossings of `clk`, i.e., the time where $\texttt{clk}_+ = \texttt{clk}_-$. The advantage of using differential clock is that at the Rx, the trigger point is well-defined (i.e., zero). In contrast, it is difficult to define an appropriate trigger point for single-ended clock at Rx due to problems such as impedance mismatch and DC wandering which make the steady state values of 0 and 1 at Rx hard to know in advance. In this homework, the Rx waveforms of `vo`, $\texttt{clk}_+$, and $\texttt{clk}_-$ are provided to you in a csv file. Your task is to process the waveforms and plot the statistical eye, for both the fixed-rate and triggered versions.



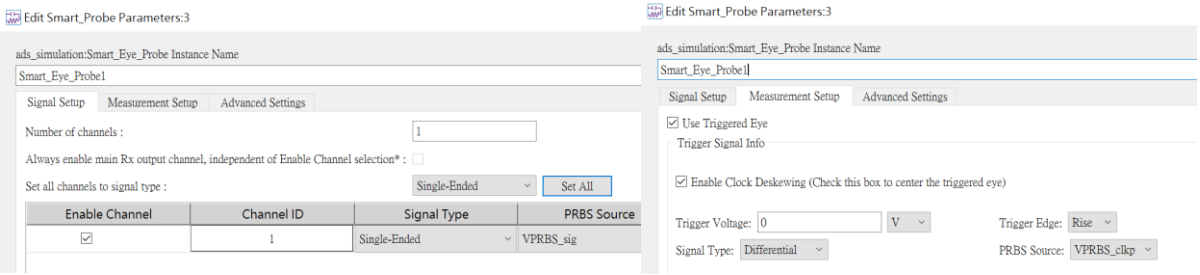
Details:

- The nominal bit rate is 40 Gbps, i.e., Tb (UI) = 25 ps. However, the actual clock rate has a 2 MHz oscillation with a period of 1 us. Namely, the clock rate constantly oscillates between 39.998 and 40.002 GHz, and the cycle time of this clock rate oscillation is 1 us. (This is an example of spread spectrum clock (SSC), with spreading rate 4 MHz / 40 GHz = 0.01% and modulation rate 1 MHz. It is worth noting that typical spreading rate in practical applications like USB is 5000ppm = 0.5%. Hence, the 0.01% spreading rate in this homework is in fact quite small. Even so, such spreading suffices to demonstrate the problem of fixed-rate eye diagram.)
- For the fixed-rate eye, all you need is the `vo` waveform (and the nominal clock rate). The clock waveform is not needed.
- For the triggered eye, you need to first evaluate $\texttt{clk} = \texttt{clk}_+ - \texttt{clk}_-$. Once the `clk` signal is obtained, you need to trace the waveform to find out the zero-crossing points. You can trigger at positive-zero-crossings only; in such case, for each trigger a 2-UI eye is constructed. You can extract the 2-UI waveform *after* the trigger, *before* the trigger, or 1-UI after and 1-UI before; it's up to you, and the resulting eyes will not differ significantly. You can also trigger at both the positive and negative zero-crossings; in such case, a 1-UI eye is constructed for each trigger.

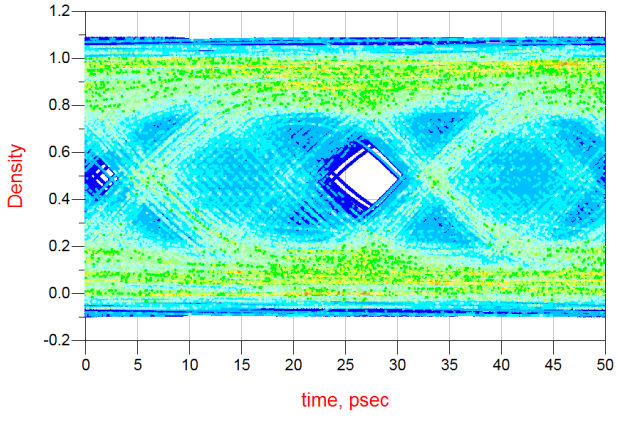
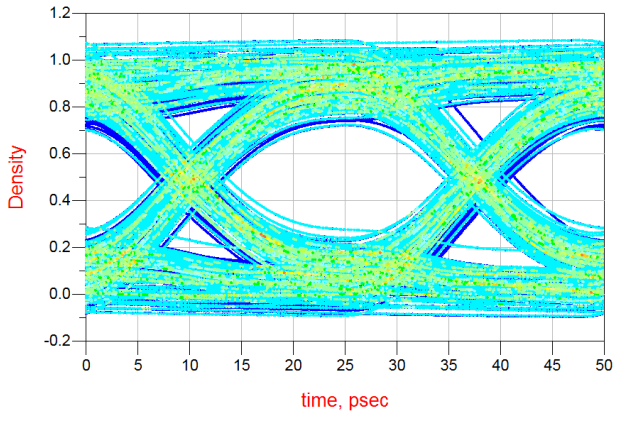
Appendix 1: The waveforms are simulated by ADS as follows:



The SSC is generated by feeding the clock generators (VtPRBS sources) a single-frequency FM source (VtSFFM). The total frequency spreading is roughly ±2 MHz. The triggered eye can be observed in ADS using the Smart_Eye_Probe, with the following settings:

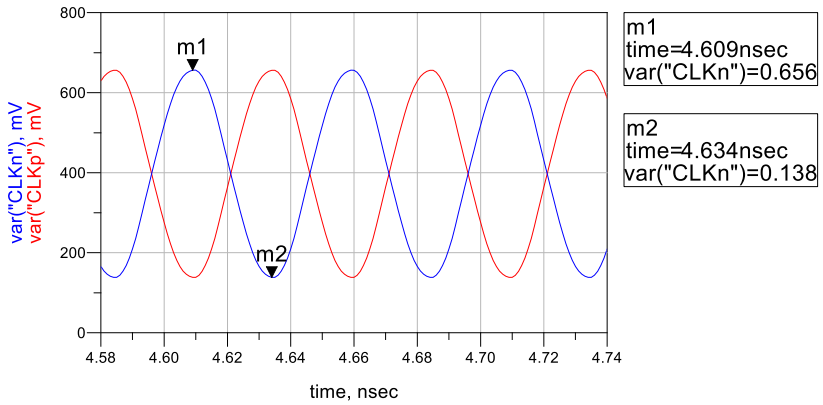


ADS's statistical eyes are as follows. Your answers should be similar (except horizontal shift and color).

Fixed-rate eye (uncheck "Use Triggered Eye")	Triggered eye
	

You are encouraged to run the ADS simulation yourself (not a required item for this HW).

For this circuit, the channel is a 20-cm microstrip line and nothing else. There is no DC wandering problem in this simple circuit. However, the source and load impedances are different. This impedance mismatch can shift the steady state values at Rx. The RX clocks have the following waveform:



It can be seen that, due to impedance mismatch at Tx and Rx, the peak value of the clock is 656 mV, while the valley is 138 mV. Therefore, for single-ended clock we may trigger at 397 mV. These values can be calculated if we know the source and load impedances. In practice, however, the Tx and Rx chips may be fabricated by different companies, and the values of the impedances may not be known exactly. This illustrates a practical difficulty, and this is why differential clock is useful in which the trigger point (0V) is independent of source/load impedances.